

FusionCore: A 23-State UKF for IMU, Encoder, GPS, and VSLAM Fusion in ROS 2: Quantitative Evaluation on Twelve Full-Length Sequences

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Abstract—FusionCore is an open-source ROS 2 sensor fusion package that fuses IMU, wheel encoder odometry, GPS, and Visual SLAM pose into a 100 Hz odometry stream using a 23-state Unscented Kalman Filter. The 23 states include gyroscope bias, accelerometer bias, and a wheel encoder yaw rate bias estimated online through GPS cross-covariance. We report results on twelve full-length NCLT sequences (55–92 min each, ~940 min total) against `robot_localization` EKF and UKF. FusionCore achieves lower Absolute Trajectory Error on ten of twelve sequences, with improvements from 1.2× to 22.2×. The `robot_localization` UKF diverges numerically on all twelve sequences.

Index Terms—sensor fusion, unscented Kalman filter, ROS 2, GPS, IMU, state estimation

I. INTRODUCTION AND MOTIVATION

Accurate pose estimation for ground robots requires fusing high-rate IMU data with low-rate GPS and wheel odometry. `robot_localization` [2] is the standard ROS solution, but several architectural choices limit practical performance: gyroscope and accelerometer biases are not filter states; GPS requires a separate `navsat_transform` node using UTM projection (subject to zone boundary failures); outlier rejection uses scalar Mahalanobis thresholds that do not account for measurement dimensionality; and noise covariance is fixed at configuration time.

FusionCore addresses each limitation in a single ROS 2 lifecycle node with the following contributions: (1) a 23-state UKF with continuous online estimation of gyroscope bias, accelerometer bias, and wheel encoder yaw rate bias; (2) ECEF-native GPS fusion eliminating UTM projection; (3) per-sensor chi-squared outlier gating calibrated per degrees of freedom; (4) automatic noise covariance adaptation from the innovation sequence; (5) ZUPT for stationary periods; (6) IMU ring-buffer retrodiction for GPS/VSLAM delay compensation; and (7) Visual SLAM pose fusion with automatic map-reinitialization recovery.

II. FILTER DESIGN

A. State Vector

The 23-dimensional state is: $\mathbf{x} = [\mathbf{p}, \mathbf{q}, \mathbf{v}, \boldsymbol{\omega}, \mathbf{a}, \mathbf{b}_g, \mathbf{b}_a, b_{ewz}]^\top$, where $\mathbf{p} \in \mathbb{R}^3$ is ENU position, $\mathbf{q}$ is unit quaternion, $\mathbf{v}, \boldsymbol{\omega}, \mathbf{a} \in \mathbb{R}^3$ are body-frame velocity, angular velocity, and acceleration, $\mathbf{b}_g, \mathbf{b}_a \in \mathbb{R}^3$ are

gyroscope and accelerometer biases, and $b_{ewz} \in \mathbb{R}$ is the wheel encoder angular velocity Z bias.

The 23rd state is the key novelty. In differential drive robots, b_{ewz} arises from wheel radius mismatch and mechanical asymmetry. FusionCore identifies it online through GPS heading cross-covariance: when GPS is available, observed heading error back-propagates into b_{ewz} . During GPS blackouts (coast mode), the estimated bias is subtracted from encoder ω_z before integration, mirroring exactly how $\mathbf{b}_g$ handles gyroscope drift.

B. Key Design Choices

Chi-squared outlier gating. Every measurement is gated by $d^2 = \boldsymbol{\nu}^\top S^{-1} \boldsymbol{\nu} \leq \tau_s$, where τ_s is set per sensor per degrees of freedom: GPS 3-DOF $\tau = 16.27$ [$\chi^2(3, 0.999)$], VSLAM 6-DOF $\tau = 22.46$, heading 1-DOF $\tau = 10.83$. `robot_localization` exposes a single scalar threshold compared against $\sqrt{d^2}$, conflating dimensionality: calibrating for a 3-DOF sensor leaves a 1-DOF sensor under-protected.

Adaptive noise covariance. Sensor noise R adapts online: $R \leftarrow (1 - \alpha)R + \alpha \hat{C}$, where $\hat{C}$ is the empirical innovation covariance from a 50-sample window and $\alpha = 0.01$. A floor $R_{ii} \geq R_{0,ii}$ prevents collapse. This is the primary mechanism through which FusionCore handles the GPS covariance mismatch described below.

III. EVALUATION

A. Setup

We evaluate on the NCLT dataset [1]: Microstrain 3DM-GX3-45 IMU (100 Hz), Segway wheel odometry (100 Hz), Novatel SPAN-CPT GPS (5 Hz). All twelve available sequences are run to full length (55–92 min, ~940 min total). Metric is ATE RMSE (meters) computed with EVO [3] against RTK ground truth. FusionCore uses a single `nclt_fusioncore.yaml` across all twelve sequences with no per-sequence tuning. `robot_localization` EKF is configured with chi-squared-equivalent thresholds: `odom0_twist_rejection_threshold: 4.03` ($\sqrt{\chi^2(3, 0.999)}$) and `odom1_pose_rejection_threshold: 3.72`.

TABLE I
ATE RMSE (METERS) ON TWELVE FULL-LENGTH NCLT SEQUENCES.

Sequence	FC	RL-EKF	RL-UKF	Winner
2012-01-08	18.6	41.2	†	FC (2.2×)
2012-02-04	49.7	265.5	†	FC (5.3×)
2012-03-31	22.0	156.5	†	FC (7.1×)
2012-05-11	9.7	11.5	†	FC (1.2×)
2012-06-15	49.2	18.2	†	RL (2.7×)
2012-08-20	98.3	10.6	†	RL (9.3×)
2012-09-28	10.8	55.7	†	FC (5.2×)
2012-10-28	29.9	60.0	†	FC (2.0×)
2012-11-04	60.1	122.0	†	FC (2.0×)
2012-12-01	21.0	90.7	†	FC (4.3×)
2013-02-23	59.4	82.2	†	FC (1.4×)
2013-04-05	12.1	268.9	†	FC (22.2×)

Numerical divergence (NaN) within first 30 s on all sequences.

B. Results

C. The GPS Covariance Mismatch Problem

The NCLT NavSatFix stream reports $\text{var}_{xy} = 9$ (3 m sigma), matching the Novatel SPAN-CPT open-sky specification. Measured against RTK ground truth, the actual p95 error ranges from 9.7 m to 53.1 m across sequences: 2 to 18 times the stated sigma. RL-EKF trusts the stated covariance and calibrates its chi-squared gate accordingly. When actual errors reach 40–200 m, those fixes land far outside the gate and get rejected. With GPS effectively disabled, RL-EKF reverts to wheel-encoder dead-reckoning, producing drift rates of 31.84 m/km and 50.11 m/km on the two worst sequences, diagnostic of open-loop operation across full runs.

FusionCore’s adaptive noise window inflates the GPS noise model in real time from the innovation sequence, recalibrating the chi-squared gate to actual error levels. When GPS actually is bad (2012-08-20 adversarial cluster, analyzed below), the gate still rejects it. The filter no longer permanently discards a sensor whose manufacturer-reported sigma is optimistic.

D. Analysis of the Two FusionCore Losses

2012-06-15 (FC 49.2 m, RL 18.2 m). The dataset’s GPS-sparsest sequence: 12,399 mode-3 fixes vs. 17,000–22,000 on all others, with a 462-second (7.7-min) blackout. During coast mode, FusionCore dead-reckons on encoder and IMU with b_{ewz} subtracted. Any residual b_{ewz} error compounds over 7.7 min: at 100 Hz, even a small heading rate residual accumulates quadratic lateral error. RL-EKF wins through a structural advantage: its two-dimensional mode has fewer divergence degrees of freedom, not because of better fusion. The architecturally correct fix is magnetometer integration providing absolute heading during GPS absence.

2012-08-20 (FC 98.3 m, RL 10.6 m). The GPS stream contains 105 mode-3 fixes located 720–840 m from RTK ground truth, concentrated in a 24-second window at the end of a 211-second blackout. These appear valid in the raw data stream. Coast mode slightly relaxes the chi-squared gate to re-acquire GPS after blackout; the tight cluster of adversarial fixes each individually pass the gate under elevated

position uncertainty and collectively pull the estimate. RL-EKF rejects them because its miscalibrated gate (which causes 10 other sequence losses) incidentally also rejects these. The fix is a velocity-consistency pre-check upstream of chi-squared gating: a fix 720 m from the dead-reckoned position after 211 s implies 3400 m/s travel speed. A `max_implied_speed` check (e.g., 20 m/s) handles this with zero cost to normal operation.

IV. CONCLUSION

FusionCore is a 23-state ROS 2 UKF fusing IMU, wheel encoders, GPS, and VSLAM at 100 Hz in a single lifecycle node. On twelve full-length NCLT sequences, it outperforms `robot_localization` EKF on 10 of 12, with gains from 1.2× to 22.2×. The RL UKF diverges numerically on all twelve. The two FC losses are attributable to diagnosable causes with identified fixes. FusionCore is available at <https://github.com/manankharwar/fusioncore> (Apache 2.0).

REFERENCES

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